

RITIK SAXENA

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PROFESSIONAL SUMMARY

Data Scientist with 3+ years of end-to-end experience across ML, NLP, Generative AI, and MLOps. Generated **€1.5M+** in monthly incremental revenue by building and deploying production ML systems at TVS Motor Company. Skilled at converting business problems into deployable AI solutions real-time APIs, NLP pipelines, multi-agent RAG systems, and cloud deployments. Completing an MSc in Data Science at Trinity College Dublin, with dissertation research on Transformer KV-cache optimisation for LLM inference. Targeting Data Scientist, ML Engineer, and AI/GenAI Engineer roles across Ireland and Europe.

EDUCATION

MSc Data Science

Sept 2025 - Aug 2026

[Trinity College Dublin](#) - Dublin, Ireland

Dissertation: Statistical Evaluation of Transformer Key-Value Caching Strategies with a Prototype Block-Based KV Engine.

Modules: Advanced Machine Learning, NLP, Statistical Data Science, Scalable Computing.

BE Computer Engineering

Jul 2018 - Aug 2022

[Thapar Institute of Engineering & Technology](#) - Punjab, India

TECHNICAL SKILLS

Languages	Python (proficient), SQL (proficient), PySpark (proficient), C++ (familiar)
ML / Deep Learning	TensorFlow, PyTorch, Scikit-learn, SciPy, Hugging Face Transformers, NLTK, SpaCy
GenAI / LLMs	LangChain, LangGraph, RAG, Prompt Engineering, Fine-tuning, vLLM, Gemini, GPT-4o, BERT, LLaMA
MLOps / Deployment	MLflow, FastAPI, Docker, Evidently, CI/CD Pipelines, REST APIs, Statistical Modelling, A/B Testing
Cloud & Databases	GCP, Vertex AI, Cloud Run, BigQuery, Azure, Databricks, App Service, CosmosDB, MongoDB, Vector DBs
Developer Tools	Git, GitHub Actions, SHAP, Pandas, NumPy, Plotly, Streamlit, VS Code

PROFESSIONAL EXPERIENCE

Data Scientist II

Oct 2024 – Aug 2025

[TVS Motor Company](#) - Bengaluru, India

ServiceMate AI - Agentic RAG Assistant for Dealer Service Workflows

LangGraph · LangChain · GPT-4o · Databricks Vector Store · Azure CosmosDB · Chainlit · Azure App Service

- Built an **agentic RAG assistant** for dealership service workers, grounding answers across 5+ knowledge sources (service manuals, SOPs, telematics, expert records).
- Designed a **7-step LangGraph workflow** covering query rephrasing, brand extraction, classification, vector DB routing, and grounded response generation.
- Integrated **2 retrieval backends** (Databricks Vector Store + Azure CosmosDB) to route diagnostic and procedural queries to the most relevant source.
- Added **4 hallucination-control flows**: retrieve-first generation, unrelated-query blocking, weak-context fallback, and answer grounding evaluation.
- Deployed via **Chainlit on Azure App Service**, providing a conversational interface without building a custom frontend.

Fine-Tuning BERT for Domain-Specific NLP

BERT · Hugging Face Transformers · PyTorch · Scikit-learn

- **Sentiment Analysis:** Continued BERT pretraining (MLM) on 200 GB+ of service data; fine-tuned 3-class model achieving **96% accuracy**.
- **Named Entity Recognition:** Fine-tuned BERT-NER on 10K+ call transcripts achieving **F1 94%**, enabling **15% faster** targeted follow-ups for sales and support teams.

Data Scientist

Aug 2022 – Sept 2024

[TVS Motor Company](#) - Bengaluru, India

Lead Classification Engine

Python · Databricks · MLflow · FastAPI · Docker · Scikit-learn · SHAP

- Deployed real-time ML model predicting purchase probability, contributing to **€1.5M+ estimated monthly incremental revenue** across 2-wheeler sales.
- Scaled to **200+ dealers** across India; improved HOT lead follow-up within 2 hours from 30% to **90%**.
- Implemented full MLOps lifecycle: MLflow registry, CI/CD pipelines, REST API serving, and **SHAP-based explainability**.
- Built an integrated monitoring framework tracking model, data, system, and business KPIs for production reliability.

Website Hyper-Personalisation Engine

Python · GCP · FastAPI · Docker · XGBoost · LightGBM · Vertex AI

- Built a **multi-label ensemble classifier** (Random Forest, XGBoost, LightGBM) to predict real-time CTA propensity and personalise website journeys.
- Deployed on **GCP Cloud Run** via FastAPI containerised service; load-tested for low-latency real-time scoring at scale.
- Validated business impact via **A/B testing**; maintained production drift detection pipeline on Vertex AI.

PROJECTS

Search Engine with Automated Evaluation

[GitHub](#)

Java · Apache Lucene · Maven · GitHub Actions · Information Retrieval

- Built a full search engine in Java with document indexing, topic-based retrieval, and TREC-format ranked output using Apache Lucene.
- Automated retrieval quality evaluation via **GitHub Actions**, benchmarking MAP, P@20, and nDCG@20 across every code change.

Spatio-Temporal Audio-Video Analysis for Infant Pain Detection

Python · TensorFlow · NLP · Multimodal Deep Learning

- Developed a **multimodal deep learning system** fusing audio, visual, and textual features from video for automatic infant pain detection.
- Achieved **64% classification accuracy** on a low-resource medical dataset, comparable to reported human inter-rater agreement benchmarks.

KEY ACHIEVEMENTS

- Built and launched a **student marketplace app** for Irish university students independently, from idea to working product.
- Placed **4th of 20 and 6th of 30 teams** in Dublin-based hackathons (2025–2026) while managing full-time MSc coursework.
- Maintain a hands-on portfolio spanning production ML, GenAI systems, search engines, web apps, and LLM inference benchmarking.